A MULTI-MODAL FRAMEWORK FOR QUANTIFYING NARRATOR RELIABILITY IN 'ILM AR-RIJAL USING SEMANTIC SENTIMENT ANALYSIS AND GRAPH NEURAL NETWORKS

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1. INTRODUCTION

1.1 Overview

The authentication of prophetic traditions (Hadith) constitutes one of the most intricate scholarly endeavours in classical Islamic civilisation. Central to this process is 'Ilm ar-Rijal, i.e. the science of evaluating the men and women who form the chains of narration (*isnād*) through which a Hadith is transmitted. Within this tradition, the discipline of Jarh wa Ta'dīl (literally "criticism and commendation") provides the epistemological framework by which scholars assess the moral probity ('adālah) and mnemonic precision (dabt) of every individual narrator in a given chain (Kamali, 2022). Over centuries, this practice produced vast historical corpuses of biographical literature, monuments of scholarly precision such as Ibn Hajar al-'Asqalāni's twelve-volume Tahdhīb al-Tahdhīb each representing thousands of qualitative evaluative judgements that collectively determine the trustworthiness of the prophetic record.

In the contemporary era, the digitisation of classical texts through global repositories has made the Hadith corpus accessible on an unprecedented scale. Yet, paradoxically, this very accessibility has created a new challenge: the capacity of modern scholarship to rigorously authenticate and verify Hadith at the speed of digital dissemination has not kept pace with the volume of their circulation. Recent scholarship identifies a "validity crisis" where the speed of digital access often outpaces the ability to verify complex chains of transmission (Mosa, 2023). The manual cross-referencing of biographical entries across hundreds of classical volumes remains a laborious, expert-dependent process that is inherently resistant to scale, presenting a compounding challenge for the field as a whole (Mosa, 2023).

This project proposes to close this gap by developing a multi-modal, data-driven framework that automates the process of narrator reliability evaluation grounded in the classical principles of Jarh wa Ta'dīl. Specifically, the framework integrates three advanced computational paradigms:

1. Transformer-based Arabic Named Entity Recognition (NER) for the precise extraction and disambiguation of narrator identities from classical texts (Muther et al., 2023);
2. Graph Neural Networks (GNN) for modelling the structural and temporal integrity of isnād transmission chains (Mahmoud et al., 2024); and
3. Semantic Sentiment Quantisation to translate qualitative scholarly evaluations into a standardised, continuous reliability index (Zarog, 2023).

The integration of these techniques is informed by prior foundational work on narrator disambiguation (Mahmoud et al., 2022), structured transmission datasets (Farooqi et al., 2024), and sequence-based isnād modelling (Alshutayri et al., 2022). By uniting these techniques, this research aims to produce a reproducible, explainable, and epistemologically grounded system that directly supports the traditional science of 'Ilm ar-Rijal through the methods of modern data science.

1.2 Problem Background

The evaluation of Hadith authenticity through 'Ilm ar-Rijal remains a cornerstone of Islamic scholarship, as the credibility of a prophetic tradition is fundamentally tied to the reliability of its chain of transmission (*isnad*). This rigorous verification system, described by classical scholars as a "protective shield" for the Sunnah, necessitates that every link in the chain be scrutinised for both moral integrity (ʿadalah) and mnemonic precision (dabt) (Kamali, 2022). Traditionally, this process is a manual, labour-intensive endeavour that requires scholars to cross-reference thousands of qualitative biographical entries across vast historical corpuses, such as the 12-volume Tahdhib at-Tahdhib of Ibn Hajar or the foundational Andalusian Maliki assessments in Ibn ʿAbd al-Barr's At-Tamhid (Al-Asqalani, d. 852 AH; Ibn ʿAbd al-Barr, d. 463 AH).

Feasibility studies confirm that machine learning can assist in automating isnād analysis, provided that the nuances of classical Arabic and the methodological standards of traditional jurisprudence are preserved (Usman, 2025; Mahmoud et al., 2022). The consonantal script and orthographic ambiguity of classical texts make this particularly challenging, as minor variations in name orthography can lead to misidentification of narrators across manuscript traditions (El Ganadi et al., 2025). Consequently, there exists an urgent need for Natural Language Processing (NLP) techniques to both extract and disambiguate narrator identities, and for Graph Neural Networks (GNN) to model the complex relational networks those narrators form (Farooqi et al., 2024; Mahmoud et al., 2022).

Recent work has confirmed the technical viability of this approach: transformer-based models like AraBERT have achieved high accuracy in identifying ambiguous narrators from biographical text, while GNN frameworks can generate node embeddings to capture the relative position and credibility of a narrator within the broader transmission network (Ahmad et al., 2025; Mahmoud et al., 2022). By transitioning to a systematically quantified framework, researchers can achieve higher consistency and speed in evaluation without compromising the epistemological roots of the science (Tarmom, 2022a).

1.3 Problem Statements

Despite advancements in Arabic NLP, two primary challenges persist in the computational study of Hadith:

1. While current models have improved in Named Entity Recognition (NER) for extracting narrator names and lineages, there is a critical lack of computational frameworks capable of quantifying narrator reliability. Identifying a narrator is insufficient; the framework must be able to "judge" their standing based on historical evidence (Ahmad, 2025).
2. Current models typically treat the text and the transmission network as separate entities. They fail to integrate the semantic sentiment of *Jarh wa Ta'dil* (scholarly praise or criticism) found in biographical texts with the structural features of the transmission network. Without this multi-modal integration, automated grading remains superficial and fails to capture the latent patterns of teacher-student reliability explored in recent work by Mosa (2025).

1.4 Project Objectives

The main objectives of this project are:

1. To develop an NLP classification model that maps qualitative scholarly judgments to a standardised, quantitative reliability index (0.0 - 1.0).
2. To construct a Knowledge Graph representing teacher-student relationships and utilise GNNs to evaluate the structural integrity and temporal consistency of transmission chains.
3. To evaluate the framework's accuracy by comparing automated reliability scores against established scholarly Hadith gradings (*Sahih*, *Hasan*, *Dha'if*).

1.5 Project Scope

This project operates within a clearly defined set of boundaries to ensure that the research remains focused, technically feasible, and evaluable within the constraints of a single academic year.

1.5.1 Data Sources

The framework will be trained and tested exclusively on digitised classical Arabic biographical texts available through the OpenITI and Shamela repositories, specifically the Taqrīb al-Tahdhīb of Ibn Hajar al-Asqalāni and the Tahdhīb al-Kamāl of al-Mizzī. These texts contain the most comprehensive and widely referenced collections of narrator biographical assessments (*tarājim*) and thus represent the most suitable ground-truth source for Jarh wa Ta'dīl labels.

1.5.2 Scale and Dataset

The initial experimental scope targets approximately 1,500 narrator biographical entries for building the reliability model and 1,000 validated Hadith isnāds drawn from the *Al-Kutub al-Sittah* (The Six Canonical Books, i.e. Sahih al-Bukhari, Sahih Muslim, Sunan Abu Dawud, Sunan at-Tirmidhi, Sunan an-Nasa'i, and Sunan ibn Majah) for end-to-end testing. This scale is consistent with comparable feasibility studies in the literature (Mahmoud et al., 2022; Najeeb, 2022) and is sufficiently large to support statistically meaningful evaluation without exceeding available computational resources.

1.5.3 Technical Scope

The system will focus on three interconnected tasks:

1. Narrator extraction: Identifying proper names, kunya (honorifics), and nisba (place-based identifiers) from *isnāds*;
2. Entity disambiguation: Resolving alias variations to canonical narrator identities via a Knowledge Graph; and
3. Reliability scoring: Mapping aggregated *Jarh wa Ta'dīl* labels to a continuous index $W = [0.0 - 1.0]$. The primary programming language is Python, utilising Arabic-specific NLP libraries (e.g., CAMeL Tools, PyArabic) and transformer models (e.g., AraBERT, MARBERT).

1.5.4 Application Domain Limitations

This research is strictly concerned with the **structural evaluation of *isnād*** (*sanad* criticism). The analysis of Hadith textual content (*Matn* criticism), which requires comparative jurisprudential reasoning across the four Sunni schools, falls outside the scope of this project. Furthermore, any automated reliability score produced by the framework is intended as a **decision-support tool** for scholars, not as a definitive theological ruling or fatwa. The epistemological authority of qualified human *muhaddithun* is expressly preserved.

1.5.5 Out of Scope

The following are explicitly excluded from this project:

- *Matn* (textual body) analysis or content-based Hadith grading.
- Coverage of Shī'a *isnād* evaluation methodologies.
- Real-time or web-based deployment of the scoring pipeline.
- Languages other than Arabic (e.g. Urdu or Persian Hadith commentaries).

2. LITERATURE REVIEW

2.1 Impact of Hadith Grading

The grading of Hadith is not merely an academic exercise; it carries profound consequences for Islamic law (fiqh), theology ('aqidah), and everyday Muslim practice. Hadith that are classified as *Sahih* (sound) or *Hasan* (good) are considered legally normative and form the second most authoritative source of Islamic jurisprudence after the Quran (Kamali, 2022). Conversely, *Da'if* (weak) or *Mawdu'* (fabricated) traditions, if acted upon without qualification, risk introducing unsubstantiated practices into religious life (Azizah, 2023). The stakes of misclassification are therefore not abstract; they directly affect rulings on worship, commerce, family law, and ethics observed by over 1.9 billion Muslims globally.

The reliability of this grading system hinges entirely on the veracity of the individuals who transmitted these reports across generations. A single unreliable narrator anywhere in the chain is sufficient to downgrade a Hadith, regardless of how credible the surrounding transmitters may be (Kamali, 2022). This sensitivity to individual narrator credibility means that even small errors in evaluation, whether due to misidentification of a narrator or misapplication of a critical term, can propagate into downstream jurisprudential decisions. The cumulative impact of such errors, particularly in the rapidly expanding digital Hadith landscape, represents a serious epistemological risk that this research directly seeks to mitigate (Azizah, 2023; Nur, 2022).

Beyond the religious domain, the science of Hadith grading has drawn increasing attention from historians, linguists, and information scientists as one of the oldest and most systematic peer-review and source-criticism methodologies in human intellectual history (Usman, 2025). The methodological rigour of classical *muhaddithun* (Hadith scholars), who developed chains of trust, biographical verification, and cross-referencing centuries before modern academic citation standards, offers a compelling analogue to contemporary knowledge verification challenges (Ghriss, 2025). Applying computational tools to this tradition, therefore, carries interdisciplinary significance that extends well beyond Islamic studies alone (A'thoina & Hasballah, 2025).

2.2 Hadith Authentication Process

The classical process of authenticating a Hadith is a dual-track methodology that evaluates both the *sanad* (chain of narrators) and the *matn* (textual body) of a tradition (Kamali, 2022). Sanad criticism, the focus of this research, concerns the continuity and credibility of the transmission chain from its originator (typically the Prophet Muhammad ﷺ) to the collector who recorded it. Each narrator in the chain must be proven to have met the preceding narrator, to have possessed an accurate memory, and to have been morally upright throughout their life. Any break in the chain (*inqita'*) or identified moral failing in a transmitter is sufficient grounds to weaken or reject the report entirely (Usman, 2025).

The grading system that emerges from this process is hierarchical and multi-tiered. At the apex sits the *Sahih* ("sound") Hadith, which satisfies all criteria of chain continuity, narrator credibility, and absence of hidden defects (*'illah*) or contradictions (*shudhadh*). Below this are

degrees of *Hasan* (“good”), where minor shortcomings in narrator memory are present but do not fatally undermine the report. *Da'if* (“weak”) Hadiths fail one or more of these conditions, while *Mawdu'* (“fabricated”) traditions are those for whose narrators deliberate falsehood has been established (Kamali, 2022). This taxonomy forms the classification target for the automated scoring system proposed in this research, where the aim is to computationally approximate the output of this traditional grading process (Ahmad et al., 2025).

The process of collecting the qualitative assessments that inform these grades is documented in a vast body of biographical literature known as *kutub ar-rijal* (books on narrators). Scholars such as Yahya ibn Ma'in, Ahmad ibn Hanbal, and later Ibn Hajar al-'Asqalani dedicated their careers to interviewing, cross-referencing, and documenting the reliability status of thousands of narrators. These books contain the raw *Jarh wa Ta'dil* labels, terms of praise and criticism that this project aims to computationally extract, quantify, and aggregate into an automated reliability index (Mghari et al., 2024; A'thoina & Hasballah, 2025).

2.3 'Ilm ar-Rijal: An Expansive Biographical Study

'Ilm ar-Rijal (the science of narrators) is one of the most elaborate biographical knowledge systems ever developed in any civilisation (Anas, 2025). At its peak, classical scholars tracked the life histories, teachers, students, travel records, and credibility assessments of over 500,000 narrators, a feat of distributed, multi-generational data collection that has no direct parallel in pre-modern scholarship (Usman, 2025). The core output of this science, the biographical dictionary (*mu'jam* or *tabaqat*), catalogues narrators by generation, geographic region, or alphabetical order, and assigns each an evaluative standing based on the consensus of prior critics (Dallil et al., 2022).

The terminology employed within *'Ilm ar-Rijal* is highly technical and hierarchically structured. Terms of commendation (*ta'dil*) range from the superlative *Thiqah Thiqah* (supremely reliable) down to *Saduq* (truthful but occasionally erring), with each grade carrying different implications for the admissibility of a narrator's reports. Terms of criticism (*jarh*) range from mild reservations such as *Layyin* (“soft”, i.e., slightly weak) to severe condemnations such as *Kadhdhab* (“liar”) or *Dajjal* (“arch-deceiver”) (Kamali, 2022). The precise semantic weight of each term, and the conditions under which it strengthens or overrides competing assessments, is the subject of a dedicated sub-discipline, one that this project attempts to operationalise through Semantic Sentiment Quantisation (Mghari et al., 2024).

A central computational challenge in this science is the problem of resolving **narrator identity**. The Arabic naming convention comprises a given name (ism), patronymic (*kunya*), lineage chain (*nasab*), tribal affiliation (*nisba*), and epithet (*laqab*), which means that a single narrator may be referred to by dozens of orthographic variants across different texts and centuries (Mahmoud et al., 2022). The scholar known canonically as Abu Dawud as-Sijistani, for example, may appear across manuscripts in numerous abbreviated or dialectally inflected forms. Disambiguating these variants into a single canonical identity is a prerequisite for any reliable automated grading system, and is the primary challenge that the Knowledge Graph and NER components of this research are designed to solve (Najeeb, 2022; Mosa, 2025).

2.4 Advancements in Arabic Named Entity Recognition (NER)

Modern processing of classical Arabic text has moved decisively away from rule-based constraints toward deep bidirectional representations. Research by Mahmoud et al. (2022) using the AR-Sanad 280K dataset established a new baseline for narrator disambiguation, proving that large-scale datasets are essential for handling the morphological density of biographical entries.

The current "State of the Art" (SOTA) utilises Pretrained Language Models (PLMs). Alghamdi et al. (2025) demonstrated that Transformer-based models like AraBERT and MARBERT significantly outperform previous iterations in identifying fraudulent content and extracting technical entities. These advancements facilitate high-accuracy Entity Linking, allowing the system to resolve identities across centuries even when names are identical (Mahmoud et al., 2022; Najeeb, 2022).

2.5 Knowledge Graphs (KG) for Entity Disambiguation

A Knowledge Graph (KG) is a structured representation of real-world entities and the semantic relationships between them, encoded as a directed graph of nodes and edges (Ji et al., 2022). In the context of *'Ilm ar-Rijal*, this structure is particularly well-suited to the problem of narrator identity resolution: each node represents a canonical narrator identity, edges encode transmission relationships (teacher-student pairs), and node attributes store the accumulated biographical metadata, including *Jarh wa Ta'dil* assessments drawn from classical *kutub ar-rijal* (Dallil et al., 2022; Shafie, 2021). Unlike flat relational databases, KGs allow queries that traverse multi-hop relationships, enabling the system to infer that two orthographically different name mentions in separate manuscripts refer to the same historical individual based on their shared network neighbourhood (Aftar et al., 2024; Dolgolov & Romanova, 2023; Hogan et al., 2021).

Within the domain of digital Hadith studies, several prominent Knowledge Graphs (KGs) have been developed to model the complex interdependencies of *isnād* chains (Alam, 2021). Systems such as **NarratorsKG** encode the topological structure of narrator transmission networks, mapping tens of thousands of distinct identities alongside their diverse *kunya* and *nisba* variants (Dabeek et al., 2021). In a typical hybrid retrieval pipeline, an NER model extracts name mentions, which are then resolved against the KG's candidate pool using both local semantic context and global graph signals (El Ganadi et al., 2025). Specifically, graph-level metrics like **PageRank centrality** are utilised to weigh a narrator's historical prominence; this helps resolve ambiguities where purely lexical matching fails, a common issue in the classical corpus where a single name may refer to five or more distinct individuals (Dabeek et al., 2021; Mosa, 2025).

More recently, the integration of KGs with Transformer-based models has defined the current state of the art. Mosa (2025) proposed a KG-Transformer framework that encodes narrator node embeddings directly into the attention mechanism of a pre-trained Arabic language model, allowing the model to simultaneously process the local textual context of a name mention and its global relational position within the transmission network. This synergy between structural and semantic representations addresses what earlier approaches treated as separate problems, which are text understanding and graph reasoning, and produces entity

disambiguation accuracy that approaches or exceeds 97% on the AR-Sanad 280K-v2 benchmark (Mahmoud et al., 2024; Mosa, 2025). The construction and querying of such a KG therefore constitutes a foundational prerequisite for the GNN-based reliability analysis described in the following section.

2.6 Advancements in Graph Neural Networks (GNN) for Isnad Analysis

Traditionally, the "structural integrity" of an *isnad* was analysed using Social Network Analysis (SNA), focusing on explicit metrics like degree centrality and betweenness to visualise transmission flows (Shafie, 2026; Siddiqui et al., 2023). While useful for visualisation, SNA is limited in its ability to learn latent relational features (Apgar et al., 2023).

The field has recently evolved toward Graph Neural Networks (GNN) and Knowledge Graphs (KG). Mahmoud et al. (2024) introduced narrator identification by querying isnad graphs, while Mosa (2025) proposed a Knowledge Graph-Transformer framework to synergise structure and semantics. Unlike SNA, GNNs can represent a narrator's "reliability" as a latent feature influenced by their entire neighbourhood of teachers and students. This allows for the detection of "hidden" patterns, such as temporal inconsistencies where a student claims to have heard from a teacher they could not have met, thus automating a core task of the *muhaddithun*.

2.7 Advancements in Semantic Sentiment Quantisation

A major challenge in automating Hadith grading is translating qualitative scholarly praise (*Ta'dil*) or criticism (*Jarh*) into numerical formats. Recent feasibility studies by Ahmad et al. (2024) confirm that Machine Learning is increasingly capable of automating isnad analysis when technical labels are correctly weighted.

Current advancements focus on Semantic Sentiment Quantisation, where technical terms such as *Thiqah* (reliable), *Saduq* (truthful), and *Munkar* (rejected) are mapped to specific reliability weights. Mghari et al. (2024) introduced Narrator2Vec, which uses word embeddings to create dense representations of a narrator's standing. This allows for a continuous reliability index $W \in [0, 1]$, where the framework calculates a composite probability score for an entire chain by aggregating the individual quantised weights of each narrator. This move toward "Explainable AI" ensures that the quantitative output remains rooted in traditional Islamic epistemology (I'thoina & Hasballah, 2024). The synthesis of these three advancements, i.e. transformer-based identity resolution, GNN-driven structural analysis, and sentiment-based weight quantisation, forms the multi-modal foundation of this research.

2.8 The Research Gap: Automated Trust & Reliability Scoring

Despite the significant advancements documented in the preceding sections, a critical gap persists at the intersection of these research streams: no published study has yet delivered a complete, end-to-end pipeline that unifies narrator extraction, identity disambiguation, and probabilistic reliability scoring into a single, evaluatable system. The majority of existing work terminates at either the extraction or disambiguation stage, treating the production of a quantified trustworthiness score as a future direction rather than an achieved outcome (Ahmad et al., 2025). The feasibility of automated Hadith grading has been acknowledged, but a

concrete and reproducible implementation against established canonical benchmarks remains absent from the literature (Tarmom, 2022b).

The separation between textual NLP work and graph-structural work further compounds this gap. Studies employing AraBERT or similar transformer models for NER do not, as a rule, feed their outputs into Knowledge Graph or GNN frameworks for structural validation (Alghamdi et al., 2025). Conversely, graph-based analyses of isnād networks rarely incorporate the semantic richness of biographical *Jarh wa Ta'dil* texts as node features (Mosa, 2025; Mahmoud et al., 2024). This disciplinary fragmentation means that existing systems capture either the semantic dimension of narrator reliability or the structural dimension, but not both simultaneously. The multi-modal integration that would make automated grading genuinely robust remains a largely unexplored frontier (Ghriss, 2025).

The standardisation of evaluation metrics also presents a significant gap. While individual studies report precision, recall, or F1-scores for their specific sub-tasks, no shared benchmark protocol exists for assessing the accuracy of an end-to-end automated grading system against the ground-truth classifications found in *Al-Kutub al-Sittah* (Farooqi et al., 2024). This absence of a common evaluation framework makes it difficult to compare approaches across studies or to claim generalised progress toward the broader goal. Establishing such a benchmark protocol is therefore a secondary but important contribution of the present research (Mahmoud et al., 2022).

This project directly targets these gaps by proposing a unified pipeline that integrates transformer-based NER, Knowledge Graph entity linking, GNN structural analysis, and Semantic Sentiment Quantisation into a single framework evaluated against pre-graded Hadith from the canonical collections. In doing so, it advances the field from the current state of fragmented sub-system studies toward a reproducible, multi-modal solution grounded in both classical Islamic epistemology and contemporary machine learning practice (A'thoina & Hasballah, 2025).

3. PROJECT METHODOLOGY

3.1 Research Design

This project adopts an experimental research design underpinned by the **Design Science Research Methodology** (DSRM). The central artefact is an end-to-end automated pipeline that ingests classical Arabic biographical text, extracts and disambiguates narrator entities, and produces a continuous reliability score for each narrator, thereby operationalising the traditional science of *Jarh wa Ta'dīl* through computational means. Development proceeds through three sequential phases, each corresponding to a distinct research objective defined in Section 1.4. Each phase produces a testable intermediate output that feeds into the next, ensuring that failures can be isolated and addressed incrementally. The methodology is structured as follows:

- Phase A: Establishes the data foundation;
- Phase B: Constructs the extraction and disambiguation engine; and
- Phase C: Implements the scoring model and end-to-end evaluation.

Together, these phases form the technical pipeline illustrated in Section 3.5.

3.2 Phase A: Data Ingestion and Preprocessing

3.2.1 Data Sources

The primary data sources are twofold:

1. Biographical Texts: Digitised editions of Taqrīb al-Tahdhīb (Ibn Hajar al-'Asqalānī) and Tahdhīb al-Kamāl (al-Mizzī), sourced from the OpenITI corpus and the Shamela Digital Library. These texts provide the *Jarh wa Ta'dīl* labels (qualitative scholarly assessments) that form the ground-truth input for the reliability scoring model.
2. Benchmark *Isnād* Dataset: The AR-Sanad 280K dataset, which contains approximately 280,000 synthetically augmented *isnād* chains with annotated narrator boundaries used for training and evaluating the NER component.

3.2.2 Preprocessing Pipeline

Classical Arabic text presents unique preprocessing challenges due to its consonantal script, optional diacritical marks (*tashkīl*), and the absence of standardised punctuation in historical manuscripts. The preprocessing pipeline addresses these challenges through the following steps:

1. Encoding Normalisation: All source texts are converted to UTF-8, with Byte Order Marks (BOM) and legacy Windows-1256 artefacts stripped.
2. Unicode Normalisation: The Arabic text is normalised to NFKC form. Variant forms of Alif (Ǐ ,ᾼ ,Ἱ) are collapsed to bare Alif (ا), and Ta Marbuta (ة) is normalised to Ha (ه) where contextually appropriate, using the CAMeL Tools normalisation utilities.
3. Diacritic Stripping: Optional *tashkīl* (short vowel marks) and *shadda* are removed to produce a consonantal skeleton, reducing orthographic variance without sacrificing semantic content.

4. Tokenisation: Sentences are tokenised using the Farasa segmenter, which handles Arabic clitics (conjunctions, prepositions, and definite articles prefixed to words) by splitting them into separate tokens. This step is critical for downstream NER, as narrator names frequently appear with attached particles (e.g., عن + و → وعن, literally "and" + "from" -> "and from").
5. *Isnād–Matn* Segmentation: Each Hadith record is split into its *isnād* (chain) and matn (body) segments using transmission keyword detection. Transition markers such as حدثنا ("narrated to us"), أخبرنا ("informed us"), and عن ("from") are used as boundary signals. Only the *isnād* portion is retained for downstream processing.

3.2.3 Expected Output

The output of Phase A is a cleaned, tokenised corpus of approximately 1,500 biographical entries and 1,000 segmented isnād chains, stored in a structured JSON format with the following schema per record: `{id, raw_text, normalised_text, tokens[], source_book, entry_type}`.

3.3 Phase B: Named Entity Recognition and Knowledge Graph Construction

3.3.1 NER Model Architecture

Narrator extraction is performed using a fine-tuned Transformer-based model. The base model is **AraBERT v2**, a pre-trained bidirectional encoder trained on a large Modern Standard Arabic corpus. The model is fine-tuned on the AR-Sanad 280K dataset for a token classification task using the BIO (Beginning, Inside, Outside) tagging scheme. The following entity types are annotated:

Tag	Entity Type	Example (Arabic)	Transliteration
B-NARRATOR	Beginning of narrator name	عبد ٱلرحـ	(ʿAbd ar-Raḥmān)
I-NARRATOR	Inside narrator name	بن صخر	(bin Ṣakhr)
B-KUNYA	Teknonymic honourific	أبو هر	(Abū Hurayra)
B-NISBA	Geographic/tribal affiliation	ٱلدوسـي ٱلزهراني	(Ad-Dawsī az-Zahrānī)
O	Outside any entity	حدثنا	("We narrated")

Table 1.1: Example of a Narrator Entity

The model is trained for 10 epochs with a learning rate of 2^{e-5}, using the AdamW optimiser with a linear warmup schedule over the first 10% of training steps. Evaluation is performed on a held-out 20% test split of the AR-Sanad dataset, with **micro-averaged F1-score** as the primary metric.

3.3.2 Knowledge Graph Construction

Extracted entity mentions are linked to canonical narrator identities stored in a **Knowledge Graph (KG)**. The KG is constructed using the Python 'NetworkX' library (with optional migration to Neo4j for scalability) and populated from the biographical entries in the Tahdhīb al-Kamāl. Each node in the KG represents a unique narrator and stores the following attributes:

- Canonical Name (الاسم الكامل)
- Known Aliases (الكنى والألقاب): Alternative names, kunya, and laqab
- Birth/Death Dates (where available)
- Geographic Region (nisba-derived): E.g. if the *nisba* is Al-Bukhari, then we can assume from Bukhara, Iran, or the area around it.
- Jarh wa Ta'dīl Labels: The raw evaluative terms assigned by classical scholars

Edges represent **transmission relationships** (teacher → student) extracted from *isnād* chains. When the NER model produces a name mention, the KG is queried using a combination of string similarity (Levenshtein distance on the consonantal skeleton) and graph neighbourhood matching (shared teachers/students) to resolve the mention to its most probable canonical identity.

3.3.3 Expected Output

The output of Phase B is a populated Knowledge Graph containing approximately 1,500 canonical narrator nodes with linked aliases, biographical metadata, and transmission edges, alongside an NER model capable of extracting narrator mentions from unseen *isnād* with a target F1-score of ≥ 0.90.

3.4 Phase C: Probabilistic Trust Scoring and Evaluation

3.4.1 Semantic Sentiment Quantisation

The core innovation of this phase is the translation of qualitative *Jarh wa Ta'dīl* labels into a continuous reliability score on the interval [0.0, 1.0]. This is achieved through **Semantic Sentiment Quantisation**, which proceeds in two steps:

1. Term Mapping: Each classical evaluative term is assigned a base weight according to the hierarchical grading scale established in the classical Hadith sciences. For example:

Term (Arabic)	Transliteration	Grade	Base Weight
ثقة ثقة	Thiqah thiqah	Highest *ta'dil*	1.00
ثقة	Thiqah	Strong *ta'dil*	0.90
صدوق	Sadūq	Moderate *ta'dil*	0.70
ليّن	Layyin	Mild *jarh*	0.20
كذّاب	Kadhdhāb	Severe *jarh*	0.00

Table 1.2: Example of Term Weighting

2. Aggregation: Where multiple scholars have assessed the same narrator using different terms, the individual weights are aggregated into a single composite score. Two aggregation strategies will be compared:
    a. A **weighted arithmetic mean**, where the weight of each assessment is influenced by the historical prominence of the assessing scholar; and
    b. a **Bayesian posterior estimation,** where each successive assessment updates a prior distribution over the narrator's reliability.

3.4.2 Chain-Level Scoring

Once each narrator in a given *isnād* has been assigned an individual reliability score w_i, the chain-level reliability R is computed as the product of individual scores:

$$R(chain) = \prod w_i \text{ for all narrators } i \text{ in the chain.}$$

This multiplicative model reflects the classical principle that a chain is only as strong as its weakest link, as a single severely criticised narrator drives the entire chain score toward zero. The chain-level score R is then mapped to the traditional grading categories (*Sahih*, *Hasan*, *Da'if*, *Mawdu'*) using threshold boundaries calibrated against the ground-truth labels in the evaluation set.

3.4.3 Evaluation Protocol

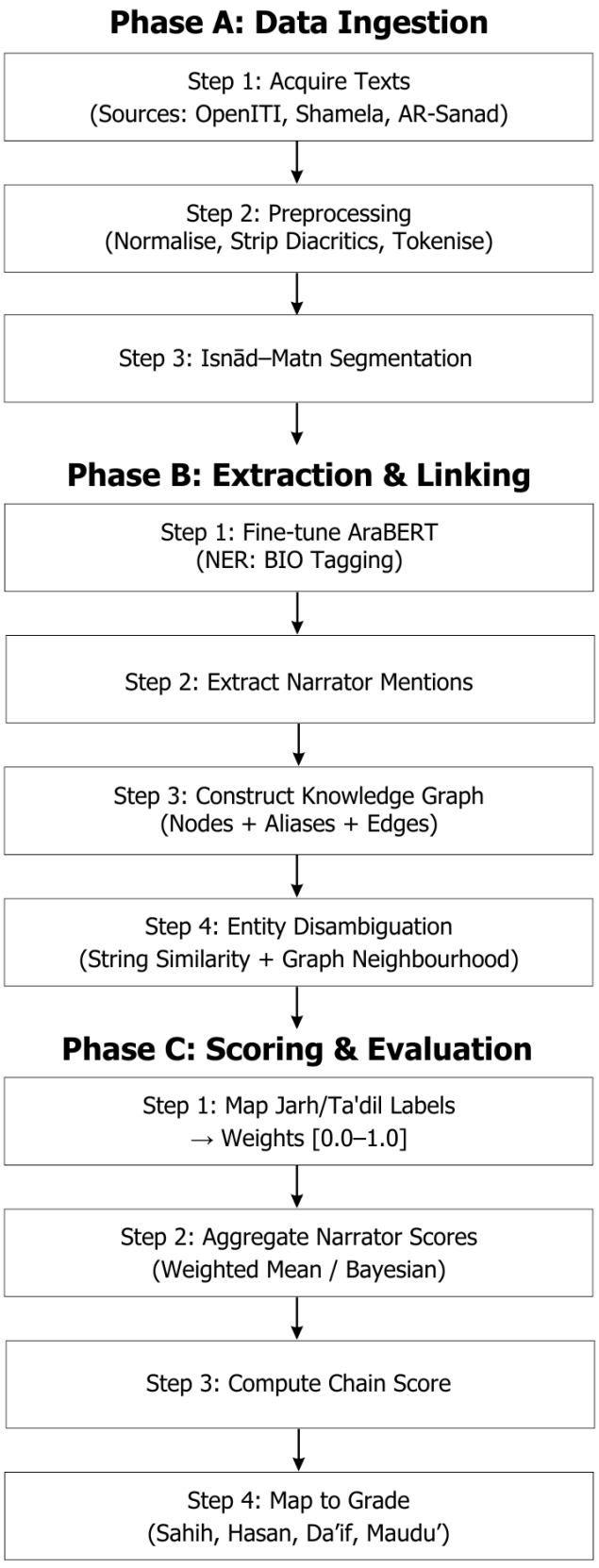
The automated grades are evaluated against **ground-truth Hadith classifications** drawn from pre-graded collections in the Al-Kutub as-Sittah (the Six Canonical Books). The following metrics are reported:

Metric	Purpose
Accuracy	Overall proportion of correctly classifi Hadiths
Precision/Recall/F1	Per-class performance for each gradi category
RMSE	Root Mean Squared Error between predict chain scores and expected ordinal grades
Cohen's Kappa	Inter-rater agreement between automated a classical human grades

Table 1.3: Purposes of Metrics Used in this Project

A baseline comparison is conducted against a naive majority-class classifier and a simple lexical lookup model (direct term-to-grade mapping without graph features) to quantify the contribution of each pipeline component.

3.5 Flow Chart of Activities



3.6 Milestones and Dates

The following table shows the weekly milestones with their deliverable respectively:

Week	Milestone	Deliverable
1-2	Project Initialisation	Finalised project summary; supervisor approval
3-4	Literature Review	Completed Chapter 2; dataset acquisition
5-6	Methodology Design	Completed Chapter 3; proposal draft
7	Proposal Defence	Oral defence and proposal submission
8-9	Phase A Implementation	Cleaned corpus; preprocessed JSON dataset
10-11	Phase B Implementation	Trained NER model; populated Knowledge Graph
12-13	Phase C Implementation	Trust scoring module; preliminary evaluation results
14	FYP1 Progress Report	Written report and presentation

3.7 Gantt Chart

```mermaid
gantt
    title FYP 1 Development Timeline
    dateFormat  YYYY-MM-DD
    axisFormat  %b %d
```

```
    section Proposal Phase
    Project Summary & Approval        :a1, 2026-03-01, 14d
    Literature Review (Ch. 2)         :a2, after a1, 14d
    Methodology Design (Ch. 3)        :a3, after a2, 7d
    Proposal Drafting & Revision      :a4, after a3, 7d
    Proposal Defence                  :milestone, 2026-04-22, 0d

    section Phase A – Data
    Data Acquisition (OpenITI, AR-Sanad) :b1, 2026-04-23, 7d
    Preprocessing Pipeline            :b2, after b1, 7d

    section Phase B – NER & KG
    AraBERT Fine-tuning (NER)         :c1, after b2, 10d
    Knowledge Graph Construction      :c2, after c1, 7d

    section Phase C – Scoring
    Sentiment Quantisation Module     :d1, after c2, 7d
    Chain Scoring & Evaluation        :d2, after d1, 7d

    section Reporting
    FYP1 Progress Report Drafting     :e1, after d2, 10d
    Final Presentation Preparation    :e2, after e1, 7d
    FYP1 Report Submission            :milestone, 2026-07-10, 0d
```

3.8 Tools and Technologies

Below are the tools, libraries and technologies used to build this project, along with their purpose for the project:

Category	Tool/Library	Purpose
Language	Python 3.10+	Primary development language
NLP Preprocessing	CAMeL Tools, PyArabic, Farasa	Arabic normalisation, tokenisation, segmentation
Transformer Model	AraBERT v2 (HuggingFace)	Fine-tuned NER for narrator extraction
Deep Learning	PyTorch, Transformers (HuggingFace)	Model training and inference
Graph Processing	NetworkX/Neo4j	Knowledge Graph construction and querying
Evaluation	scikit-learn	Classification metrics (F1, Precision, Recall, Kappa)
Data Format	JSON, CSV	Structured intermediate data storage
Version Control	Git/GitHub	Source code and experiment tracking
Documentation	Markdown, LaTeX	Proposal and report writing

5. REFERENCE

(Note: Ensure all references follow **Gaya UMS Version 2020** exactly to avoid rejection.)

1. Introduction

Azizah, N. (2023). The Digital Era of Hadith: Challenges of Authenticity and Opportunities for Innovation. Sao Jurnal IAIN Parepare.

Boukhaled, M. A. (2022). A Machine Learning based Study on Classical Arabic Authorship Identification. Proceedings of the 14th International Conference on Agents and Artificial Intelligence, 489–495. https://doi.org/10.5220/0010969100003116

Dallil, S., et al. (2022). Development of a Normalized Hadith Narrator Encyclopedia with TEI. SciELO México. https://www.scielo.org.mx/scielo.php?script=sci_arttext&pid=S1405-55462022000301283

El Ganadi, A., Gagliardelli, L., Aftar, S., & Ruozzi, F. (2025). Digital Maktaba Project: Proposing a Metadata-Driven Framework for Arabic Library Digitization. CEUR Workshop Proceedings. https://ceur-ws.org/Vol-3937/short13.pdf

Farooqi, A. M., Malick, R. A. S., Shaikh, M. S., & Akhunzada, A. (2024). Multi-IsnadSet MIS for Sahih Muslim Hadith with chain of narrators, based on multiple ISNAD. Data in Brief, 54, 110439. https://doi.org/10.1016/j.dib.2024.110439

Kamali, M. H. (2022). A Textbook of Hadith Studies: Authenticity, Compilation, Classification and Criticism of Hadith. Islamic Foundation.

Nur, A. (2022). Digital Hadith: Preservation and Documentation in the Electronic Age.

Pavlova, V. (2025). Multi-stage Training of Bilingual Islamic LLM for Neural Passage Retrieval. arXiv. https://doi.org/10.48550/arxiv.2501.10175

Tarmom, A. (2022). Compression versus Machine Learning for Classifying Modern Arabic Code-Switching in Social Media and Classical Arabic Hadith [Doctoral dissertation, University of Sheffield]. White Rose eTheses Online.

Usman, A. (2025). Islamic Scholarship in the Modern Era: Contributions, Critiques and Innovations. UIS Publishing.

2. Literature review

Ahmad, N., Ahmed, S., & Abdullah, M. (2025). Hadith grading & machine learning: Feasibility of automatic isnād-analysis. Al-Jamei Research Journal, 3(2), 155–170. https://aljamei.com/index.php/ajrj/article/view/155

Anas, M. (2025). Hadith narrators' biographies and their role in the reliability of transmission. *Contemporary Journal of Social Science Review*, *3*(3), 139–155. https://doi.org/10.63878/cjssr.v3i3.1131

Mahmoud, S., Nabil, E., Saif, O., & Torki, M. (2024). Narrator identification by querying Sanad graph and utilizing the NarratorsKG on AR-Sanad 280K-v2 dataset. Neural Computing and Applications. Advance online publication. https://doi.org/10.1007/s00521-024-10194-2

Mahmoud, S., Saif, O., Nabil, E., Abdeen, M. A. R., ElNainay, M., & Torki, M. (2022). AR-Sanad 280K: A novel 280K artificial sanads dataset for hadith narrator disambiguation. Information, 13(2), 55. https://doi.org/10.3390/info13020055

Mosa, M. A. (2025). Synergizing structure and semantics: A knowledge graph-transformer framework for narrator disambiguation in hadith networks. Digital Scholarship in the Humanities. Advance online publication. https://doi.org/10.1093/llc/fqaf088

Najeeb, M. M. (2022). A Hidden Markov Model-based tagging approach for Arabic isnads of hadiths. Mathematical Problems in Engineering, 2022, Article 7160509. https://doi.org/10.1155/2022/7160509

Shafie, O. (2021). KASHAF: A knowledge-graphs approach search-engine for hadith analysis & flow-visualization [Doctoral dissertation, ProQuest Dissertations & Theses]. https://search.proquest.com/docview/0e10e1179ecd610c2032df2193e3e795

Feasibility & Advancements References (APA 7)

Alghamdi, J., Albukhari, A., & Al-Dala'in, T. (2025). Pretrained models against traditional machine learning for detecting fake hadith. Electronics, 14(17), 3484. https://doi.org/10.3390/electronics14173484

A'thoina, I., & Hasballah, F. (2025). Bridging revelation and algorithm: Integrating artificial intelligence and malakah in the epistemology of Ḥadīth criticism. Digital Muslim Review, 3(1), 38–52. http://digitalmuslimreview.or.id/index.php/dmr/article/view/38

Ghriss, A. (2025). Crowdsourced information authentication: A graph-based model from the science of hadith. OpenReview. https://openreview.net/forum?id=F7CgAEPvAv

Mghari, M., Bouras, O., & Hibaoui, A. E. (2024). Narrator2Vec: An efficient narrator representation in hadith literature using word embedding. Arabian Journal for Science and Engineering. Advance online publication. https://doi.org/10.1007/s13369-023-08224-7

Mohamed Emad Elden, F. (2025). Isnad AI at IslamicEval 2025: A rule-based system for identifying religious texts in LLM outputs. *Proceedings of The Third Arabic Natural Language Processing Conference: Shared Tasks*, 540–559. https://doi.org/10.18653/v1/2025.arabicnlp-sharedtasks.74